Topological Spaces

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1 Topologies

1.1 Definitions

We denote $\mathcal{P}(X)$ as the power set of X.

Definition. (Topology) Let X be a set. A *topology* on X is a collection of sets $T \subseteq \mathcal{P}(X)$ such that

(i) $\emptyset, X \in T$,
(ii) T is closed under (possibly uncountable) unions.
(iii) T is closed under finite intersections.

A set X with a topology T is called a *topological space* of X. An element of X is called a *point* and elements of T are called *open sets*. If $x \in U \in T$ we say U is an open neighbourhood of x. Strictly we should always denote (X, T) for a topological space, but when T is clear, we just write X for the topological space.

Definition. (Continuity) If (X, T_X) and (Y, T_Y) are topological spaces then a function $f : X \to Y$ is called *continuous* if for $U \in T_Y$, $f^{-1}(U) \in T_X$.

Definition. (Homeomorphism) A function $f : (X, T_X) \to (Y, T_Y)$ is a *homeomorphism* if it is continuous and has a continuous inverse.

Definition. If $T \subseteq T'$ are topologies on X then we say that T is *coarser* and T' is *finer*. The identity function $d : (X, T) \to (X, T')$ is continuous.

1.2 Topologies from metrics

If (X, d) is a metric space, recall that a subset $U \subseteq X$ is called *open* if for every point $x \in U$ there exists a $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U$.

Proposition. If T_d is the subset of X which are open under the metric d, then (X, T_d) is a topological space. We will call this the topology on X induced by the metric d.

Proof. Tautologically we have that $\emptyset \in T_d$. Clearly we have that $X \in T_d$ too. Let $\{U_\alpha\}_{\alpha \in I}$ be a collection of open sets in T_d with a (possibly uncountable) index set I. Let

$$x \in \bigcup_{\alpha \in I} U_\alpha.$$

Then $x \in U_\beta$ for some $\beta \in I$, so U_β is open hence there exists a $\varepsilon > 0$ such that $B_\varepsilon \subseteq U_\beta \subseteq \bigcup_{\alpha \in I} U_\alpha$, hence $\bigcup_{\alpha \in I} U_\alpha$ is open.

Now suppoes that I is finite, and $x \in \bigcap_{\alpha \in I} U_\alpha$. For each α there exists a $\varepsilon_\alpha > 0$ such that $B_{\varepsilon_\alpha}(x) \subseteq U_\alpha$. Take $\varepsilon = \inf_{\alpha \in I} \varepsilon_\alpha$, so $B_\varepsilon(x) \subseteq B_{\varepsilon_\alpha}(x) \subseteq U_\alpha$ for all α, hence we have that $B_\varepsilon(x) \subseteq \bigcap_{\alpha \in I} U_\alpha$ so it's open. Hence T is a topology. $\square$

Now we have lots of examples we can use for topological spaces. For example we have that topology induced by the Euclidean metric on $\mathbb{R}^d$ which we will call the Euclidean topology. For any $X \subseteq \mathbb{R}^d$ we can have a topology induced by the Euclidean metric too, like $\mathbb{Q}$, $[0,1]$, $(0,1)$.

Proposition. If we have two metric spaces $(X, d_X), (Y, d_Y)$ and we have $f : X \to Y$, the f is continuous in the metric space sense if and only if it is continuous in the topological space sense (with the topologies induced by the metric d_X and d_Y respectively).

Proof. Let $f : X \to Y$ be continuous in the metric space sense. Let U be an open set in T_{d_Y} so we need to show that $f^{-1}(U)$ is open. Let $x \in f^{-1}(U)$, so $f(x) \in U$. Hence there exists an $\varepsilon > 0$ such that $B_\varepsilon(f(x)) \subseteq U$. So since f is continuous there exists a $\delta > 0$ such that if $d_X(x, x') < \delta$, then $d_Y(f(x), f(x')) < \varepsilon$. Hence $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. So $B_\delta(x) \in f^{-1}(U)$, hence $f^{-1}(U)$ is open.

Now let's do the converse and suppose that $f : X \to Y$ is continuous in the topological sense. Fix some $x \in X$ and $\varepsilon > 0$. Consider $B_\varepsilon(f(x))$ which is open in Y. Then $f^{-1}(B_\varepsilon(f(x)))$ is in T_{d_X}. It contains x so there exists a $\delta > 0$ such that $x \in B_\delta(x)$, so

$$d_X(x, x') < \delta \implies d_Y(f(x), f(x')) < \varepsilon.$$

So f is continuous in the metric sense. $\square$

Definition. Let (X, T) be a topological space and $x_1, x_2, \dots \in X$ say. We say that x_n convergences to x if for every open neighbourhood U of x there exists a N such that $x_n \in U$ for all $n \geq N$.

Proposition. If (X, d) is a metric space with topology T_d then a sequence (x_n) converges in the metric sense if and only if it converges in the topological sense.

Proof. Suppose it convergens in the metric sense to x. Then for all $\varepsilon > 0$ there exists a N such that for all $n \geq N$ we have that $x_n \in B_\varepsilon(x)$. If U is a neighourhood of x then there there is some ε such that the ball of radius ε centred at x is contained in U. Conversely if (x_n) converges in the topological sense to x, let $\varepsilon > 0$ and consider the open ball centred at x with radius ε. Now $B_\varepsilon(x)$ is an open neighbourhood of x so there exists an integer N such that $x_n \in B_\varepsilon(x)$ for all $n > N$. Hence (x_n) converges to x in the metric sense. $\square$

Consider $\mathbb{R}$ and $(0,1)$ with the Euclidean metric and topology. Then the two spaces are related, by the function $(0,1) \to \mathbb{R}$ by $\tan^{-1} x$ which is invertible. Hence we say the two spaces are homeomorphic, and $\mathbb{R} \cong (0,1)$. However the two spaces are not isometric since $\mathbb{R}$ is not complete under the Euclidean metric and $(0,1)$ is not. Hence the property of completeness is not a topological property: it is a property induced by the metric.

Definition. (Discrete topology) Let X be a set. The *discrete* topology is the topology $T_{\text{discrete}} = \mathcal{P}(X)$ (so every set is open).

Remark. Any function from (X, T_{discrete}) to any space is continuous. This toplogy can be induced by the discrete metric, where $d(x,y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$. So $B_{\frac{1}{2}}(x) = \{x\}$ so $\{x\}$ is open, hence all

the sets are open.

Definition. (Indiscrete topology) Let X be a set. The *indiscrete* topology $T_{\text{indiscrete}} = \{\emptyset, X\}$ (as little as possible sets are open).

Remark. A function from any space to $(X, T_{\text{indiscrete}})$ is continuous. This topology does not come from a metric unless X is a singleton set. This is because if $x \neq y$ then $d(x, y) = \varepsilon > 0$, so $y \notin B_\varepsilon(x)$ and since y is arbitrary, then $B_\varepsilon(x) = \{x\} = X$.

Let $X = \{o, c\}$. Then let $T = \{\emptyset, \{o, c\}, \{o\}\}$ be a topology of X. This is called the Sierpinski space. It as the property that every sequence converges to c. A continuous function $f : T \to (X, T_{\text{Sierpinski}})$ is exactly an open subset of Y.

Let $X = \mathbb{R}$ we'll define the right order topology on X as

$$T_{\text{ord}} = \{(a, \infty) \mid -\infty \leq a \leq \infty\}.$$

Let $\{(a, \infty)\}_{a \in I}$ be a collect of elements of T_{ord}. Then

$$\bigcup_{a \in I} (a, \infty) = (\inf_{a \in I} a, \infty) \in T_{\text{ord}}.$$

Similarly for finite I,

$$\bigcap_{a \in I} (a, \infty) = (\max_{a \in I} a, \infty) \in T_{\text{ord}}$$

1.3 Bases and subbases

Definition. (Basis) Let T be a topology of X. A *basis*, $B \subseteq T$ for T is a subcollection such that every element of T is a union of elements in B.

Definition. (Subbasis) Let T be a topology of X. A *subbasis*, $S \subseteq T$ for T is a subcollection such that every element of T is a union of sets which are finite intersections of elements of S.

Lemma. Let $f : (X, T_X) \to (Y, T_Y)$ and $S \subseteq T_Y$ is a subbasis. If $f^{-1}(U)$ is open for all $U \in S$ then f is continuous.

Proof. If $V \subseteq T_Y$, then $V = \bigcup_{a \in I} V_a$ where $V_a \in \bigcap_{b \in J_a} U_{a,b}$ with $U_{a,b} \in S$ and J_a finite. Then

$$f^{-1}(V) = \bigcup_{a \in I} V_a = \bigcup_{a \in I} \left(\bigcap_{b \in J_a} f^{-1}(U_{a,b}) \right) \in T_X,$$

by the axioms of the topology. □

Consider the Euclidean topology on $\mathbb{R}^n$. The collection $B = \{B_r(x) \mid x \in \mathbb{R}^n, r > 0\}$ is a basis. Likewise the collection of n-cubes everywhere are also a basis. Interestingly the set $QB \subseteq B$ with balls at rational points with rational radii is also a basis. This is interesting since QB is countable while B is uncountable and $\mathcal{P}(\mathbb{R}^n)$ is $\aleph_2$.

Definition. (Closed set) Let (X, T) be a topological space. A subset $C \subseteq X$ is *closed* if $X \setminus C \in T$.

Proposition. Let (X, T) be a topological space and $\mathcal{F} = \{C \subseteq X \mid C \text{ closed}\}$. Then
(i) $\emptyset, X \in \mathcal{F}$;
(ii) $\mathcal{F}$ is closed under (possibly uncountable) intersections;
(iii) $\mathcal{F}$ is closed under finite unions.

Proposition. A function $f : X \to Y$ between topological spaces is continuous if and only if the preimage of every closed set is closed.

Definition. Let (X, T) be a topological space. Let $A \subseteq X$ be a subset of X. Then
(i) The closure $\bar{A}$ is the smallest (by inclusion) closed set containing A so

$$\bar{A} = \bigcap_{S \text{ closed}, A \subseteq S} S.$$

(ii) We say that A is dense in X if $A = \bar{A}$.
(iii) The interior $\mathring{A}$ is the largest open set contained in A so

$$\mathring{A} = \bigcup_{S \text{ open}, S \subseteq A} S.$$

Definition. (Limit point) Let X be a topological space and $A \subseteq X$. A *limit point* of A is a point in X which is a limit of a sequence in A.

Proposition. If C is a closed subset of (X, T), then the limit points of C lie in C.

Proof. Let $\{x_n\}$ be a sequence in C with limit x_∞. If $x_\infty \notin C$, then $x_\infty \in X \setminus C$ which is open. Then if $x_n \to x_\infty$ then we should have that $x_n \in X \setminus C$ for $n \geq N$ but $x_n \in C$ so $x_n \notin X \setminus C$ which is a contradiction. $\square$

Corollary. A limit point of a A lies in $\bar{A}$.

For an example $\overline{\mathbb{Q}} = \mathbb{R}$ since any real number is a limit of a sequence of rational numbers. We have that $\overline{(0,1)} = [0,1]$ too. The cocountable topology on $\mathbb{R}$ is the topology $T_{\text{countable}} = \{\emptyset\} \cup \{\mathbb{R} \setminus C \mid C \text{ countable}\}$ Let $\{x_n\}$ be a sequence in $\mathbb{R}$, for $x \in \mathbb{R}$ consider $\{x\} \cup \{\mathbb{R} - \{x_n\}\}$ is open and contains x. If $x_n \to x$, then x_n must be in a U for all $n \geq \mathbb{N}$ so $x_n = x$ for all $n \geq N$. Hence the convergent sequences are exactly the eventually constant sequences with the limits being the value they are eventually constant to. So the limit points of a set A are A under this topology. However almost all A is not closed. For example $(0,1)$ is not closed since $\mathbb{R} \setminus (0,1)$ is not countable. But the closure of $(0,1)$ must be closed, so it must be $\mathbb{R}$ hence the sense of limit points and closure are actually two very different properties in topology instead of metric spaces.

1.4 Hausdorff spaces

Definition. (Hausdorff) A space (X, T) is *Hausdorff* if for $x \neq y \in X$ there are open neighbourhoods $x \in U$, $y \in V$ with $U \cap V = \emptyset$.

Remark. This is the notion that points are seperated by open sets.

Lemma. If the topology T is induced by a metric then it is Hausdorff.

Proof. If $x \neq y$ then $d(x, y) = s > 0$. So consider $U = B_{s/2}(x)$ and $V = B_{s/2}(y)$. The triangle inequality shows that $U \cap V = \emptyset$ and we know all balls are open. □

Proposition. If a space is Hausdorff then a sequence in X has at most 1 limit.

Proof. Let (x_n) be a sequence in X. Suppose it has limits $y \neq z \in X$. Let U and V be disjoint local neighbourhoods sfor y and z respectively. Then $x_n \in U$ for all $n \geq N_1$ and $x_n \in V$ for all $n \geq N_2$. So if we take that $N = \max\{N_1, N_2\}$ then for all $n \geq N$, we have that $x_n \in U \cap V$ which is empty, hence we have a contradiction. □

Proposition. If (X, T) is Hausorff then points are closed.

Proof. Let $x \in X$. We want to show that $\{x\} = \overline{\{x\}}$. Let $y \neq x$. Let U, V be disjoint neighbourhoods of x and y respectively. We know that $x \in X \setminus V$ which is closed. Hence $\overline{\{x\}} \subseteq X \setminus V$. But $y \notin V$, so y is not in the closure of $\{x\}$ hence the closure of $\{x\}$ is just $\{x\}$, so $\{x\}$ is closed. □

Let's see an example. Let X be an infinite set and consider the cofinite topology on X. Take two non-empty open sets, so

$$(X \setminus F) \cap (X \setminus F') = X \setminus (F \cup F')$$

which is non-empty since $F \cup F'$ is finite and X is infinite so the set on the RHS is non-empty hence the space is not Hausdorff.

1.5 Defining new topologies on existing ones

We have three main ways to define new topologies when given a topology already.

1.5.1 The subspace topology

Definition. (Subset topology) Let (X, T_X) be a topological space. Let $Y \subseteq X$ a subset. The *subset topology* on Y is

$$T\,|_Y = \{Y \cap U \mid U \in T\}.$$

Definition. (Subspace) A subspace of (X, T) is a subset equipped with the subspace topology.

Proposition. The subset topology is a topology.

Proof. Simple exercise of the axioms. □

Proposition. The inclusion map $\iota : (Y, T\mid_Y) \to (X, T)$ is continuous. In fact $T\mid_Y$ is the constant topology on Y such that the inclusion map is continuous.

Proof. Let $U \in T$ then $\iota^{-1}(U) = U \cap Y \in T\mid_Y$ by definition. So it is continuous. Suppose $\iota : (Y, T') \to (X, T)$ is continuous. For $U \in T$, $\iota^{-1}(U) \in T'$ so $T\mid_Y \subseteq T'$. □

A further point of view, a function $f : (z, T_z) \to (Y, T\mid_Y)$ is continuous if and only if $\iota \circ f$ is continuous.

Lemma. (Gluing Lemma) Let $f : X \to Y$ be a function between topological spaces.

(i) If $\{U_\alpha\}_{\alpha\in I}$ are open subsets which cover X and each $f\mid_{U_\alpha} : U_\alpha \to Y$ are continuous (where U_α is given the subspace topology) then f is continuous.

(ii) If $\{C_\alpha\}_{\alpha\in I}$ is a finite collection of closed sets containing X and $f\mid_{C_\alpha} : C_\alpha \to Y$ is continuous for each $a \in I$ then f is continuous.

Proof. Let $V \subseteq Y$ be open. We want to show that $f^{-1}(V)$ is open. We know that

$$\begin{aligned} f^{-1}(V) = ({}^{-1}V) \cap X = f^{-1}(V) \cap \left(\bigcup_{\alpha\in I} U_\alpha \right) \\ = \bigcup_{\alpha\in I} f^{-1}(V) \cap U_\alpha \end{aligned}$$

Since $f\mid_{U_\alpha}$ are continuous, we have that $f^{-1}\mid_{U_\alpha}$ is open in U_α in the subspace topology. So there exists a W open in X such that $f^{-1}\mid_{U_\alpha}(V) = U_\alpha \cap W$ hence this is the intersection on open subsets of X so is open in X, hence since the union of open subsets is open $f^{-1}(V)$ is open, so f continuous.

The second part can be proved the same using the closed set definition of continuity. □

If (X, d) is a metric space with topology T_d and $Y \subseteq X$ then $T_d\mid_Y$ is the topology induced by $d\mid_Y$.

1.5.2 The quotient topology

Definition. (Quotient topology) Let (X, T_X) be a topological space, $\sim$ an equivalence relation on X and $X/\sim$ is the set of equivalence classes, and $\pi : X \to X/\sim$ the equivalence map. The *quotient toplogy* on $X/\sim$ is

$$T_{X/\sim} = \{U \subset X/\sim \mid \pi^{-1}(U) \in T_X\}.$$

Proposition. $T_{X/\sim}$ is indeed a topology.

Proof. $\emptyset = \pi^{-1}(\emptyset) \in T_X$ so $\emptyset \in T_{X/\sim}$. $X = \pi^{-1}(X/\sim) \in T_X$ so $X/\sim \in T_{X/\sim}$. Let $\{U_\alpha\}$ be a collection of sets of $T_{X/\sim}$, then

$$\pi^{-1}\left(\bigcup_{\alpha\in I} U_\alpha\right) = \bigcup_{\alpha\in I} \pi^{-1}(U_\alpha),$$

and $\pi^{-1}(U_\alpha) \in T_X$, so the union is too. Hence $\bigcup_{\alpha\in I} U_\alpha \in T_{X/\sim}$. We have a similar proof for finite intersections. □

Proposition. The quotient map $\pi : (X, T_X) \to (X/\sim, T_{X/\sim})$ is continuous and $T_{X/\sim}$ is the finest topology for which this is true.

Proof. This is a tautology. □

An alternative characterisation of the quotient topology is that $f : X/\sim \to Y$ is continuous if and only if $f \circ \pi : X \to Y$ is continuous.

Definition. For a continuous function $g : (X, T_X) \to (Y, T_Y)$ is a *quotient map* if it surjective and $U \in T_Y \iff g^{-1}(Y) \in T_X$.
Given, this construct $\sim$ on X by $x \sim x' \iff g(x) = g(x')$. There is an induced function $G : X/\sim \to Y$ sending $G([x]) = g(x)$.

Remark. This function G is a bijection and continuous with a continuous inverse. This means that G is a homeomorphism, so $X/\sim \cong Y$.

Let's see an example on $\mathbb{R}$. Consider $x \sim y \iff x - y \in \mathbb{Z}$. What is $\mathbb{R}/\sim$? Consider $f : \mathbb{R} \to \mathbb{R}^2$ defined by $x \to (\sin(2\pi x), \cos(2\pi x))$. This is a continuous map so $f : \mathbb{R} \to S^1 \subseteq \mathbb{R}^2$ is also continuous and surjective. By periodicity $x \sim y \iff f(x) = f(y)$, so we get $F : \mathbb{R}/\sim \to S^1$ which we can check is a homeomorphism.

Now take the example $X = \mathbb{R} \times \{0, 1\} \subseteq \mathbb{R}^2$ with the standard subspace topology. Let $(x, i) \sim (y, j) \iff (x, i) = (y, j)$ or $x = y \neq 0$. We can then think of $X/\sim$ is a line with two origins. We cannot draw $X/\sim$ since it is not Hausdorff. Any neighbourhood of $[(0, 0)]_\sim$ intersects any neighbourhood of $[(1, 0)]_\sim$ so not Hausdorff. Hence it is not subspace of any Euclidean space.

1.5.3 The product topology

For sets X, Y the projections functions are

$$\begin{aligned} \pi_X : & X \times Y \to X \\ & (x, y) \to x \end{aligned}$$

and

$$\begin{aligned} \pi_Y : & X \times Y \to Y \\ & (x, y) \to y \end{aligned}$$

Definition. (Product topology) Let (X, T_X) and (Y, T_Y) be topological spaces. Then *product topology* on $X \times Y$ consists of open sets $U \subseteq X \times Y$ such that for $(x, y) \in U$ there

is a $V \in T_X$ and $W \in T_Y$ such that $(x, y) \in V \times W \in U$.

Proposition. This indeed is a topology and the sets $V \times W$ are a basis for $T_{X \times Y}$.

Proof. Tautologically, we have that $\emptyset \in T_{X \times Y}$. Taking $V = X, W = Y$ we have that $X \times Y \in T_{X \times Y}$. For a collection $\{U_\alpha\}_{\alpha \in I}$ of elements of $T_{X \times Y}$, let $(x, y) \in \bigcup_{\alpha \in I} U_\alpha$. Then $(x, y) \in U_\beta$ for some $\beta \in I$ so there exists neighbourhoods of x, y with their product a subset of $U_\beta \subseteq \bigcup_{\alpha \in I} U_\alpha \in T_{X \times Y}$. If I is finite and $(x, y) \in \bigcap_{\alpha \in I} U_\alpha$. Then $(x, y) \in V_\alpha \times W_\alpha \subseteq U_\alpha$ for each $\alpha \in I$. So $(x, y) \in (\bigcap_\alpha V_\alpha) \times (\bigcap_\alpha W_\alpha) \in \bigcap_\alpha U_\alpha$ and since these intersections are finite, these intersections are open. $\square$

Proposition. The projection maps

$$\pi_X : (X \times Y, T_{X \times Y}) \to (X, T_X) \qquad \pi_Y : (X \times Y, T_{X \times Y}) \to (Y, T_Y)$$

are continuous and $T_{X \times Y}$ is the coarsest topology for which this is true.

Proof. Let $V \in T_X$. Then $\pi_X^{-1}(V) = V \times Y$, so this is open. Hence π_X, π_Y are continuous.

Suppose that T' is a topology on $X \times Y$ such that π_X and π_Y are continuous, then $\pi_X^{-1}(V) = V \times Y$ is open and $\pi_Y^{-1}(W) = X \times W$ is open. So $V \times W$ is open in T', so $T_{X \times Y} \subseteq T'$. $\square$

The universal property of the product topology is that the function

$$f : (Z, T_Z) \to (X \times Y, T_{X \times Y})$$

is continuous if and only if $\pi_X \circ f : (Z, T_Z) \to (X, T_X)$ and $\pi_Y \circ f : (Z, T_Z) \to (Y, T_Y)$ are continuous. Equivalently f is componentwise continuous if and only if it is componentwise continuous.

2 Connectivity

2.1 Connected and disconnected

We know from IA Analysis I, if $f : [0, 1] \to \mathbb{R}$ is continuous, $f(0) < 0 < f(1)$ then $f(t) = 0$ for some $t \in [0, 1]$. This is a statement about continuous functions, but also about the interval $[0, 1]$. For example if we change the interval to $[0, \frac{1}{2}) \cup (\frac{1}{2}, 1]$ then this does not satisfy the intermediate value theorem. The property of the interval we're using is connectedness.

Definition. (Disconnected) A topological space X is *disconnected* if $X = U \cup V$ for U, V disjoint nonempty open sets.

Definition. (Connected) A topological space is *connected* if it is not disconnected.

If $X = U \cup V$ is disconnected, then U and V are both open and also both closed.

Any set with the coarse topology is connected, due to the lack of non-trivial open sets. A set with the discrete topology is disconnected, if it has more than 1 point, since every set is open, so the result is trivial.

The set $X = [0, \frac{1}{2}) \cup (\frac{1}{2}, 1] \subseteq \mathbb{R}$ is disconnected since $[0, \frac{1}{2})$ is open in X and $(\frac{1}{2}, 1]$ is open in X too. They are disjoint, hence X is disconnected.

Proposition. A space X is disconnected if and only if, there is a continuous surjection $f : X \to \{0, 1\}$ where $\{0, 1\}$ is equipped with the discrete topology.

Proof. Suppose that X is disconnected. So $X = U \cup V$ disjoint. The define f such that

$$f(x) = \begin{cases} 0 & x \in U \\ 1 & x \in V \end{cases}.$$

This is well-defined since U and V are disjoint. Since U and V are non-empty, the function is surjective. The preimage of $\{0\}$ and $\{1\}$ are U and V respectively which we know is open. And the preimage of $\{0, 1\}$ and $\emptyset$ are clearly open, so f is continuous.

Conversely suppose that f is continuous. Then define $U = f^{-1}(\{0\})$ and $V = \int f(\{1\})$. So since f is continuous, U and V are open. Clearly U and V is disjoint and non-empty since f is surjective. We have that $X = U \cup V$ since $X = f^{-1}(\{0, 1\}) = f^{-1}(0) \cup f^{-1}(1) = U \cup V$. □

Theorem. The spaces $[0, 1]$, $[0, 1)$, $(0, 1)$ are all connected.

Proof. Let's just consider $[0, 1]$, the rest of the proves are similar. If it was disconnected, then there is a continuous surjection

$$f : [0, 1] \to \{0, 1\} \subseteq \mathbb{R}.$$

Then

$$f(\cdot) - \frac{1}{2} : [0, 1] \to \mathbb{R}$$

is continuous and takes the values $\pm\frac{1}{2}$ only. By the intermediate value theorem, we should have that f takes the value 0 which is a contradiction hence $[0, 1]$ is connected. □

Theorem. (Generalised intermediate value theorem) Let X be a connected topological space and $f : X \to \mathbb{R}$ continuous. If there exists $x_0, x_1 \in X$ such that $f(x_0) < 0 < f(x_1)$ then there exists a $x_2 \in X$ such that $f(x_2) = 0$.

Proof. Consider the open sets $U = f^{-1}((-\infty, 0)), V = f^{-1}((0, \infty))$. f is continuous, so U, V are open. We know that x_0, x_1 exist hence U, V are non-empty. If $f(x)$ is never zero, then $X = U \cup V$ disjoint and open so X is disconnected. But X is connected hence $f^{-1}(0)$ is non-empty, so pick $x_2 \in f^{-1}(0)$, so $f(x_2) = 0$. □

Proposition. Let $f : X \to Y$ be a continuous surjection. Then X connected implies that Y is connected.

Proof. Let's show the contrapositive. Suppose that Y is disconnected. Then we have some $h : Y \to \{0,1\}$ continuous and surjective. So

$$h \circ f : X \to \{0,1\}$$

is also continuous and surjective, hence X is disconnceted. $\square$

Corollary. If X is connected and $f : X \to Y$ is continuous then $\mathrm{im}(f)$ is connected.

Proof. Apply the proposition to $f : X \to \mathrm{im}\, f$.

For example if X is a connected space and $\sim$ is an equivalence relation then $\pi : X \to X/\sim$ is a continuous surjection so $X/\sim$ is connected.

Lemma. If $f : X \to Y$ is a homeomorphism and $Z \subseteq X$, then $f\mid_Z : Z \to \mathrm{im}(f\mid_Z)$ is a homeomorphism.

Proof. Obvious. $\square$

Let's use this to show that $[0,1]$ is not homeomorphic to $(0,1)$. Suppose they are. So we have a homeomorphism $f : [0,1] \to (0,1)$. Let's now restrict f to $(0,1]$. Then by the lemma we know that $f\mid_{(0,1]}$ is a homeomorphism with

$$f\mid_{(0,1]} : (0,1] \to (0,1) \setminus \{f(0)\}$$

for some $0 < f(0) < 1$. But $(0,1]$ is connected and $(0,1) \setminus \{f(0)\} = (0, f(0)) \cup (f(0), 1)$ so $(0,1) \setminus \{f(0)\}$ is disconnected which is a contradiction.

We can do a similar process to show that S^1 is not homeomorphic to $\mathbb{R}$. We know that S^1 is connected since it is a quotient space of $\mathbb{R}$ and $\mathbb{R}$ is connected since $\mathbb{R} \cong (0,1)$. Suppose that S^1 is homeomorphic to $\mathbb{R}$. Then remove the point $(1,0) \in S^1$ and consider the restricted homeomorphism between the new spaces. $\mathbb{R}$ is no longer connected since $\mathbb{R} \setminus \{f(1,0)\} = (-\infty, f(1,0)) \cup (f(1,0), \infty)$, but $S^1 \setminus \{f(1,0)\}$ is connected since it's homeomorphic to $(0,1)$.

Proposition. Let $\{X_\alpha\}_{\alpha \in I}$ be a collection of subspaces of X. Suppose that each X_α is connected and $\bigcap_{\alpha \in I} X_\alpha \neq \emptyset$. Then $\bigcup_{\alpha \in I} X_\alpha$ is connected.

Proof. Let X be the union of all the sets. If it were disconnected then $X = U \cup V$ with U, V open, so $U \cap X_\alpha, V \cap X_\alpha$ are disjoint open subsets covering X_α. Since X_α is connected one of them must be zero so $X_\alpha \subseteq U$ or $X_\alpha \subseteq V$. This holds for each X_α but as $\bigcap_{\alpha \in I} X_\alpha \neq \emptyset$ they are all in U or all in V hence X is U or X is V so one of them is empty. $\square$

Corollary. If X and Y are connected then so is $X \times Y$.

Proof. Suppose X, Y both non-empty (since the empty set is connected). Choose $x \in X$. Consider

$$C_y = \{x\} \times Y \cup X \times \{y\}.$$

The sets $\{x\} \times Y$ and $X \times \{y\}$ intersect in (x, y) and pieces are connected by assumption so C_y is connected. Now observe that

$$X \times Y = \bigcup_{y \in Y} C_y$$

the intersection of all of C_y is $\{x\} \times Y$ which is non-empty hence the proposition applies and $X \times Y$ is connected.

2.2 Path-connectedness

Definition. (Path) If X is a topological space and $x_0, x_1 \in X$ a *path* between them is a continuous $\gamma : [0,1] \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$.

Definition. (Path connected) A topological space is *path connected* if for any two points in the space, there is a path between them.

The spaces $(0,1), [0,1], (0,1]$ are all path-connected just by taking the line $\gamma(t) = (1-t)x_0 + tx_1$. For the same reason $\mathbb{R}^n$ is path-connected and any *convex* subset $X \subseteq \mathbb{R}^n$ is too.

Now let's let $X = \mathbb{R}^2 \setminus \{0\}$ be the punctured plane. We do the same again if our linear path doesn't go through the (missing) origin. If it does, just take a circular path instead.

Proposition. A path-connected space is connected.

We'll prove the contrapositive. If X is not connected, we have a continuous surjective function $f : X \to \{0,1\}$ Let x_0, x_1 be such that $f(x_0) = 0, f(x_1) = 1$ and suppose that X is path connected so there is a path connecting x_0, x_1. So $f \circ \gamma$ is a surjective and continuous map from $[0,1] \to \{0,1\}$ hence $[0,1]$ is disconnected which is a contradiciton. $\square$

We can now show that $\mathbb{R}^n \ncong \mathbb{R}$ for $n > 1$. If it were then $\mathbb{R}^n \setminus \{0\} \cong \mathbb{R} \setminus \{0\}$. But the RHS is disconnected and the LHS is path-connected, contradiction.

Proposition. If X and Y are path-connected then $X \times Y$ is path-connected.

Proof. Omitted.

2.2.1 Path components

Let X be a space. Define an equivalence relation on X, $\sim$, defined by

$$x \sim y \iff \text{there exists a path from } x \text{ to } y.$$

Lemma. $\sim$ is indeed an equivalence relation.

Proof.

(i) $x \sim x$ since we can take the path $\gamma(t) = x$.

(ii) If $x \sim y$ then we have a path $\gamma(t)$ connecting x and y. Then we can take the path $\gamma'(t) = \gamma(1-t)$ which goes from y to x. Hence $y \sim x$.

(iii) If $x \sim y$ and $y \sim z$, there there is a path γ' connecting x to y and γ' connecting y to z. Define

$$(\gamma' \cdot \gamma)(t) = \begin{cases} \gamma(2t) & 0 \le t \le \frac{1}{2} \\ \gamma'(2t-1) & \frac{1}{2} \le t \le 1 \end{cases}.$$

This is a well defined function since $\gamma(1) = \gamma'(0)$. It is continuous by the gluing lemma applied to $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ so it is a path between x and z, so $x \sim z$.

Hence $\sim$ is an equivalence relation. $\square$

Definition. (Path components) The *path components* of X are the equivalence classes of $\sim$.

Claim. Each component is path-connected.

Proof. Suppose that x, y are in the same path component, so there exists a path γ connecting x and y. Since $\gamma\mid_{[0,t]}: [0, t] \to X$ is a path from $\gamma(0)$ to $\gamma(t$ every point on the path is in the path component, hence the path lies in the equivalence class. $\square$

2.2.2 Connected components

For a space X, define $\approx$ by $x \approx y$ if and only if there exists a subset $C \subseteq X$ connected with $x, y \in C$.

Lemma. $\approx$ is an equivalence relation.

Proof.

(i) $\{x\}$ is connected, so $x \approx x$.

(ii) Definition is symmetric so $x \approx y \iff y \approx x$.

(iii) If $x \approx y$ and $y \approx z$ then there are connected subsets C_1 containing x, y and C_2 containing y, z. Then let $C = C_1 \cup C_2$ which is connected since the intersection contains y so non-empty and contains x, z so $x \approx z$.

Hence $\approx$ is an equivalence relation. $\square$

Definition. (Connected components) Let the *connected components* of X are the equivalence classes of $\approx$.

Proposition. The connected components are connected.

Proof. Let $C \subseteq X$ be a connected component. Suppose that $f : C \to \{0, 1\}$ is a surjective continuous function. Let $f(x_0) = 0$ and $f(x_1) = 1$. As $x_0 \approx x_1$, there exists a connected space $D \subseteq X$ with $x_0, x_1 \in D$. If $d \in D$, then $d \approx x_0$, hence $D \subseteq C$. But then $f\mid_D: D \to \{0, 1\}$ is a continuous surjective function so D is disconnected, contradiction, hence C is connected. $\square$

For example $X = (-\infty, 0) \cup (-, \infty)$ has connected components $(-\infty, 0), (0, \infty)$ which are also the path-components.

For $\mathbb{Q} \subset \mathbb{R}$ the connected compoments are the singletons, so are the path-components.

Let's look at the *Topologist's sine curve*, which is defined as

$$S = \left\{ \left(x, \sin\left(\frac{1}{x}\right) \right) \in \mathbb{R} \mid 0 < x \leq 1 \right\}.$$

This is not closed and it's closure is

$$\overline{S} = S \cup (\{0\} \times [-1, 1]).$$

We want to show that $\overline{S}$ is connected but *not* path connected. Since S is the image of $(0, 1]$ under $x \to \left(x, \sin\left(\frac{1}{x}\right)\right)$ hence S is connected.

Lemma. The closure of a connected subspace is connected.

Proof. Let $C \subset X$ be a connected subspace and suppose that $\overline{C} = U \cup V$ is disconnected, so U, V are non-empty, disjoint open sets of $\overline{C}$. Thus U, V are also closed in $\overline{C}$. Hence they are closed in X too. Consider $C \cap U$ and $C \cap V$. This is a cover of C by disjoint open subsets. So one of them is empty, so $C \subset U$ without loss of generality. But since U is closed in X, $\overline{C} \subset U$ too, so $\overline{C} \cap V = \emptyset$. This contradicts the fact that $U \cup V$ was a disconnection. $\square$

Hence we have that $\overline{S}$ is connected. Note that $\{0\} \times [-1, 1]$ and S are both path connected, so to show that $\overline{S}$ is not path connected we need to show that there is no $\gamma : [0, 1] \to \overline{S}$ with $\gamma(0) \in \{0\} \times [-1, 1]$ and $\gamma(1) \in S$. Suppose that γ is such a path. As $\{0\} \times [-1, 1]$ is closed, $\gamma^{-1}(\{0\} \times [-1, 1])$ is closed, so it contains its supremum t. Then $\gamma \mid_{[t,1]}$ is a new path with $\gamma(t) \in \{0\} \times [-1, 1]$ and $\gamma(s < t) \in S$. SO we replace γ by $\gamma \mid_{[t,1]}$. So we can write $\gamma(s) = (x(s), y(s))$ for $x, y : [0, 1] \to \mathbb{R}$ continuous. For $s > 0$ we have that $y(s) = \sin\left(\frac{1}{x(s)}\right)$. Our goal is to find a sequence $s_n \to 0$ in $[0, 1]$ such that $y(s_n) = (-1)^n$. Then this is a contradiction since it should converge to $y(0)$ since y is continuous. For each n choose a $0 = x(0) < a < x\left(\frac{1}{n}\right)$ such that $\sin\left(\frac{1}{u}\right) = (-1)^n$. By the intermediate value theorem we have that $u = x(s_n)$ for some $0 < s_n < \frac{1}{n}$. So $y(s_n) = (-1)^n$ as required. $\square$

2.3 Compactness

Definition. (Cover) A collection $\mathcal{X} \subset P(X)$ is a *cover of* X if for each $x \in X$ there is a $S \in \mathcal{X}$ with $x \in S$.

Definition. (Open cover) An *open cover* of X is a cover consisting of open sets.

Definition. (Subcover) A *subcover* of $\mathcal{X}$ is a $\mathcal{X}' \subseteq \mathcal{X}$ which is also a cover.

Let's give a definition now which generalised the idea of being closed and bounded without need of a metric.

Definition. (Compact) A topological space X is *compact* if every open cover has a finite subcover.

$\mathbb{R}$ is not compact by considering the open sets $\{(n-1, n+1) \subset \mathbb{R} \mid n \in \mathbb{Z}\}$ which clearly is an open cover but has no finite subcover since there isn't even a subcover, since each $n \in \mathbb{Z}$ is only contained in exactly one open interval in the cover.

If X is a topological space with finitely many points then it is compact.

If $X = \{0\} \cup \{\frac{1}{n}\}$ for $n = 1, 2, \dots$. Then X is compact. Let $\mathcal{U}$ be an open cover of X. The point $0 \in X$ lies in some $U_0 \in \mathcal{U}$. As this is open, there exists an $\varepsilon > 0$ such that $0 \in B_\varepsilon(0) = (-\varepsilon, \varepsilon) \subset U_0$. Thus $\frac{1}{n} \in U_0$ for $0 < \frac{1}{n} < \varepsilon$ for $n > \frac{1}{\varepsilon}$. The finitely many points $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{m}$ that do not satisfy this, lie in open sets $1 \in U_1, \frac{1}{2} \in U_2, \dots, \frac{1}{m} \in U_m$. Hence $\{U_i\}_{i=0}^m$ is a finite subcover.

The space $Y = \{\frac{1}{n} \mid n = 1, 2, \dots\} \subseteq \mathbb{R}$ is *not* compact. This is because it is an infinite set and the topology is discrete since $\left\{\frac{1}{n}\right\} = Y \cap \left(\frac{1}{n+1}, \frac{1}{n+1}\right)$. Hence the cover by each of the points has no subcover at all.

Theorem. $[0, 1]$ is compact.

Proof. Let $\mathcal{U}$ be an open cover of $[0, 1]$. Consider

$$A = \{a \in [0, 1] \mid \text{there is a finite } \mathcal{U}' \subset \mathcal{U} \text{ whose union contains } [0, a]\}.$$

If $0 \leq a \leq b$ then $[0, a] \subseteq [0, b]$. So if $b \in A$ then $a \in A$. A is not non-empty since $0 \in A$ since $[0, 0] = \{0\}$ has some $U \in \mathcal{U}$ containing 0. Hence by completeness A has a supremum, so set $\alpha = \sup A \in [0, 1]$. We want to show that $\alpha = 1$. Let $\alpha \in U_\alpha \in \mathcal{U}$. If $\alpha < 1$ then there is an $\varepsilon > 0$ such that $(a - \varepsilon, a + \varepsilon) \subset [0, 1] \subset U_\alpha$. So as $\alpha - \varepsilon < \alpha$, there must be a $\alpha - \varepsilon \leq a < \varepsilon$ with $a \in A$. Hence $\alpha - \varepsilon \in A$ too. But then $[0, \alpha - \varepsilon]$ has a finite cover by elements of $\mathcal{U}$. Adding U_α we have a finite subcover of $[0, \alpha + \varepsilon/2]$ hence this contradicts the fact that α is a supremum, so $\alpha = 1$ and $[0, 1]$ is compact. □

Let's look at some consequences of compactness.

Proposition. If X is compact and $C \subseteq X$ is closed, then C is compact.

Proof. Open sets in C are of the form $C \cap U_\alpha$ for U_α open in X. Suppose we have $\{C \cap T_\alpha \mid \alpha \in I\}$ which is an open cover of C. Then $\{X \setminus C\} \cup \{U_\alpha \mid \alpha \in I\}$ is an open cover X. So as X is compact, this has a finite subcover. It can be taken to have the form $\{X \setminus C\} \cap \{U_\alpha \mid \alpha \in I'\}$ for $I' \subseteq I$ finite. As these cover X, their intersections with C cover C, so $\{C \cap U_\alpha \mid \alpha \in I'\}$ is a finite subcover of C, hence C is compact. □

Proposition. If X is Hausdorff and $C \subseteq X$ is compact, then C is closed.

Proof. We will show that $U = X \setminus C$ is open. So for $x \in U$, we have to find an open $x \in U_\alpha \subseteq U$. For each $y \in C$ use Hausdorffness to find disjoint open sets $y \in V_y$ and $x \in W_y$. Then $\{C \cap V_y \mid y \in C\}$ is an open cover of C. As C is copmact, we can find $y_1, \dots, y_n \in C$ such that $\{C \cap V_y \mid i = 1, \dots, n\}$ cover C. Let

$$U_x = \bigcap_{i=1}^n W_{y_j}.$$

This contains x and is a finite intersection of open sets, so it is open. As W_{y_i} is disjoint from V_{j_i}, $\bigcap_{i=1}^n W_{y_i}$ is disjoint from $\bigcap_{i=1}^n V_{y_i} \supseteq C$, so U_α is also disjoint fro C. So $\alpha \in U_\alpha \subseteq U = X \setminus C$. Doing this for each $x \in U$ shows that U is open. □

Proposition. If X is compact and $f : X \to Y$ is continuous, then $f(X)$ is a compact subspace of Y.

Proof. We can suppose that f is surjective by replacing Y with $f(X)$. If $\mathcal{U}$ is an open cover of Y consider $f^{-1}(\mathcal{U}) = \{f^{-1}(U) \mid U \in \mathcal{U}\}$. This is an open cover of X, so by compactness it has a finite subcover $f^{-1}(U_1), \ldots, f^{-1}(U_n)$. We claim that $U_1, \ldots, U_n$ covers Y. This is because we can write

$$X = \bigcup_{i=1}^{n} f^{-1}(U_i)$$

so

$$f(X) \subseteq \bigcup_{i=1}^{n} U_i \subseteq Y,$$

hence we have

$$\bigcup_{i=1}^{n} f(U_i) = Y$$

so Y is compact. □

Corollary. If $f : X \to Y$ is a continuous bijection from a compact space X to a Hausdorff space Y, then it is a homeomorphism.

Proof. We only need to show that $f^{-1} : Y \to X$ is continuous. We will use the closed set characterisation of continuity. Let $C \subseteq X$ be closed. We want $(f^{-1})^{-1}(C) = f(C)$ closed. We know that C is closed and X is compact hence C is compact. Hence $f(C)$ is compact and by Hausdorffness, $f(C)$ is closed, so f is a homeomorphism. □

Definition. (Sequentially compact) A metric space (X, d) is *sequientially compact* if every sequence in X has a convergent subsequence.

The metric d gives a topology T_d on X.

Lemma. (Lebesgue's number lemma) Let (X, d) be sequentially compact and $\mathcal{U} \subset T_d$ be an open cover. Then there is a $\delta > 0$ such that each $B_\delta(X)$ lies inside some element of $\mathcal{U}$.

Proof. Suppose not. Then for each $n = 1, 2, \ldots$, there exists a point $xx_n \in X$ such that $B_{\frac{1}{n}}(x_n)$ is *not* contained in any element of $\mathcal{U}$. By sequentially compactness there is a convergent subsequence $x_{n_i} \to x_\infty$. Let $x_\infty \in U \in \mathcal{U}$ and $\varepsilon > 0$ be such that $B_\varepsilon(x_\infty) \subseteq U$. For all $i \gg 0$ we have that $x_{n_i} \in B_{\varepsilon/2}(x_\infty)$ and $\frac{1}{n_i} < \frac{\varepsilon}{2}$. The triangle inequality gives a contradiction from here. So there is such a δ called the Lebesgue number for the cover $\mathcal{U}$. □

Theorem. The metric space (X, d) is sequentially comapct if and only if (X, T_d) is compact.

Proof. Suppose that (X, d) is not sequentially compact. Then there is a sequence t_n in X with no convergent subsequence. i.e. for each $x \in X$ there is an open $U_x \ni x$ which contains only

finitely many (t_i). Then $\mathcal{U} = \{U_x \mid x \in X\}$ is an open cover of (X, T_d). If $\mathcal{U}$ had a finite subcover, then as each U_x contains only finitely many values that the sequence (t_i) takes. But this is a contradiction since (t_i) does not have a convergent subsequence. Hence $\mathcal{U}$ doesn't have a finite subcover, so (X, T_d) is not compact.

Suppose now that (X, d) is sequentially comapct and $\mathcal{U}$ is an open cover of (X, T_d). Let δ be a Lebesgue number for this cover by the lemma. We want to show that there is a finite set of points $A \subseteq X$ such that

$$X = \bigcup_{a \in A} B_\delta(a).$$

Suppose not. Then for every finite set A, there is a point $x \in X$ which lies at least δ away for all $a \in A$. So we can find a sequence $x_1, x_2, \ldots$ such that $d(x_i, x_j) \geq \delta$ for all $i \neq j$. This property means it has no convergent subsequence which is a contradiction since (X, d) is sequentially compact. Hence A does exist. Now each $B_\delta(a)$ lies in some $U_a \in \mathcal{U}$ since δ is a Lebesgue number, hence $\{U_a \mid a \in A\}$ is a finite subcover of $\mathcal{U}$. □

Corollary. (Heine-Borel Theorem) A subspcae $X \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Proof. From IB Analysis II a subspace of $\mathbb{R}^n$ is sequentially compact if and only if it is closed and bounded. Hence the result follows. □

Corollary. (Extreme value theorem) If X is a compact topological space and $f : X \to \mathbb{R}$ is continuous then there are $a, b \in x$ such that

$$f(a) \leq f(x) \leq f(b) \qquad \forall x \in X.$$

Proof. the set $f(X0 \subseteq \mathbb{R}$ is compact so by Heine-Borel it is closed and bounded. So it contains its supremum, $A = f(a)$ and it's infimum $B = f(b)$. So $f(X) \subseteq [A, B]$ so we're done. □

Theorem. If X and Y are compact, then the product space $X \times Y$ is compact.

Proof. We first prove the following lemma which only uses the fact that Y is compact.

Lemma. (Tube lemma) If Y is compact, $x_0 \in X$ and $\{x_0\} \times Y \subseteq W$ is an open neighbourhood then there is an open neighbourhood $U_{x_0} \ni x_0$ such that $U_{x_0} \times Y \subseteq W$.

Proof. We can cover $\{x_0\} \times Y$ by its intersection with open sets of the form $U \times V$ with $U \times V \subseteq W$. We can do this as W is open and the sets of the form $U \times B$ are a basis for the product topology. As this slice $\{x_0\} \times Y$ is compact, this cover has a finite subcover, so there are open sets

$$U_1 \times V_1, \ldots, U_n \times V_n \subseteq W$$

whose intersection with $\{x_0\} \times Y$ cover it. Let

$$U_{x_0} = \bigcap_{i=1}^{n} U_i$$

which is an open neighbourhood of x_0. To see that $U_{x_0} \times Y \subseteq W$, consider a point $(x, y) \in U_{x_0} \times Y$. Note (x_0, y) lies on the slice, so lies in some $U_i \times V_i$. So $y \in V_i$. But $x \in U_{x_0} \subseteq U_i$ so $(x, y) \in U_i \times V_i \subseteq W$. □

Now back to the theorem. To show $X \times Y$ is compact let W be an open cover of $X \times Y$. For each $x_0 \in X$, $\{x_0\} \times Y$ can be covered by (its intersection with) finitely many $W_1, \ldots, W_n \in W$. So $\{x_0\} \times Y \subseteq \bigcup_{i=1}^n W_i$. By the tube lemma there is a neighbourhood $U_{x_0} \ni x_0$ such that $U_{x_0} \times Y \subseteq \bigcup_{i=1}^n W_i$, so $U_{x_0} \times Y$ can be covered by finitely many elements of W. Now $\{U_{x_0} \mid x_0 \in X\}$ is an open cover of X, so has a finite subcover since X is comapct. Say $U_{x_1}, \ldots, U_{x_n}$ is a finite subcover. So $X \times Y$ is covered by $U_{x_1} \times Y, \ldots, U_{x_n} \times Y$. As each of these can be covered by finitely-many elements of W, so can the whole of $X \times Y$. □

Corollary. $[0, 1]^n$ is compact.

Proof. $[0, 1]$ is compact, so the result follows by induction, using the theorem. □

We can also prove Heine-Borel without using any analysis: it is a purely topological consequence.

Corollary. (Heine-Borel) A subspace $X \subseteq R^n$ is compact if and only if it is closed and bounded.

First let X be closed and bounded. Then $X \subseteq [-N, N]^n$ for some $N > 0$. By the corollary $[-N, N]^n$ is compact and X is a closed subspace, hence X is compact.

Conversely suppose that X is a compact then using Hausdorffness we have that it is closed. To see its bounded, consider the open cover $\{X \cap (-N, N)^N \mid N \in \mathbb{N}\}$. So by compactness this must have a finite subcover, and in fact looking at the cover it can be covered by a singular set, say $X \subseteq (-N, N)^n$ for some $N \in \mathbb{N}$. Hence X is bounded. So it's closed and bounded. □

2.3.1 The finite intersection property

We say a collection $\{S_\alpha \mid \alpha \in I\}$ of subsets of a fixed set S has the *finite intersection property* if every finite intersection of S_α is non-empty.

Theorem. A space X is compact if every collection of closed subsets $\{C_\alpha \mid \alpha \in I\}$ having the finite intersection property has

$$\bigcap_{\alpha \in I} C_\alpha \neq \emptyset.$$

Proof. We translate the definition of compactness to closed sets from open sets using the complement. This takes unions to intersections and covers to empty intersections. And if we then take the contrapositive i.e.

$$\bigcap_{\alpha \in I} C_\alpha = \emptyset \implies \text{some finite intersection is empty.}$$

This can be expanded on for a full proof, but it's just linguistical. □

2.3.2 The Cantor space

Define a subspace $C \subset [0,1]$ inductively. First set $C_0 = [0,1]$. We define $C_1 = [0,1] - \left(\frac{1}{3}, \frac{2}{3}\right)$, removing the open middle third. We contine this inductively, removing the middle third from each closed interval, so

$$C_n = C_{n-1} \setminus \bigcup_{k=0}^{\infty} \left(\frac{1+3k}{3^n}, \frac{2+3k}{3^n} \right).$$

We then define the *Cantor space* as

$$C = \bigcap_{k=0}^{\infty} C_n \subset [0,1].$$

We know that C_n is closed for all $n \in \mathbb{N}$, hence C is closed. Clearly bounded hence C is compact. It contains the end points of the intervals in C_n for all n.

Proposition. There is a bijection between C and $\{0,1\}^{\mathbb{N}}$. So C has the cardinality of $\mathbb{R}$.

Proof. Let $\{0,1\} = \{\ell, r\}$. A sequence of ℓ's and r's tells us how to chose one of the intervals in each C_n by choosing the left or right third. By the finite intersection property of compactness and th efact that $[0,1]$ is compact looking at the collecction of closed sets $I_0 \supset I_1 \supset I_2 \supset \dots$. All the finite intersections of these are some I_j so non-empty hence

$$\bigcap_{n=0}^{\infty} I_n \neq \emptyset$$

by compactness. Suppose $x, y \in \bigcap_{n=0}^{\infty} I_n$. Then $x, y \in I_n$ and interval of length $\frac{1}{3^n}$. Taking $n \to \infty$ we must have that $x = y$. This defines a function

$$\phi : \{\ell, r\}^{\mathbb{N}} \to \mathbb{C}.$$

We've just shown this is well-defined and an injection. This is tautologically surjective since each sequence defines a point in C. So ϕ is a bijection. $\square$

In fact it's easy to see that ϕ^{-1} is continuous so it's a homeomorphism.

Proposition. The connected components of C are the singletons.

If $x, y \in C$ with $x < y$. We want to find a real number $x < z < y$ with $z \notin C$. Suppose not. Then $[x, y] \subset C \in C_n$ for all n. But the lengths of $C_n \to 0$ as $n\infty$, hence we can always pick n large enough so this can't happen. Hence such a z is not in C. Then $C \cap (-\infty, x)$, $C \cap (z, \infty)$ disconnects C with x in the left set and y in the right set. So x and y cannot lie in a connected subspace of C. Hence all the connected components are singletons. $\square$

3 Manifolds

3.1 Definitions and examples

Definition. (Locally Euclidean) We say that a topogical space is *locally Euclidean* (with dimension d) if for all $x \in X$ there exists an open neighbourhood of x homeomorphic to an open subset of $\mathbb{R}^d$.

Definition. (Manifold) A topological space X is a d-dimensional *manifold* if it is Hausdorff and it is locally Euclidean with dimension d.

Remark. Any open $y \in U \subset \mathbb{R}^d$ has a $y \in B_2(y) \subset U$ and $B_2(y) \cong \mathbb{R}^d$ so equivalently we could change open subset of $\mathbb{R}^d$ to a ball in $\mathbb{R}^d$.

Remark. Manifolds are also usually taken to be *second countable* (they have a countable basis). This gives lots of useful properties, but none of which we will use in this course, so we will just ignore this.

Example. Any open $U \subset \mathbb{R}^d$ is a d-dimensional manifold trivally.

Example. The sphere $S^d = \{v \in \mathbb{R}^{d+1} \mid ||v|| = 1\}$ is a d-dimensional manifold. Clearly it's Hausdorff since it's a subspace of $\mathbb{R}^d$. To see that it's locally Euclidean consider the sterographic projection of the sphere with north pole p to the plane given by

$$\begin{aligned} S^d \setminus \{p\} &\longrightarrow \mathbb{R}^d \\ q &\longrightarrow \text{the intersection of the line through } p \text{ and } q \text{ with the plane } \{0\} \times \mathbb{R}^d. \end{aligned}$$

This gives a Euclidean neighbourhood of any point except p, but we can do the same with p by just rotating the sphere.

Example. In particular the circle S^1 is a 1-dimensional manifold. It can be described in other ways too. The relation $x \sim y \iff x - y \in$ on $\mathbb{R}$ has quotient space $\mathbb{R}/\sim = \mathbb{R}/\mathbb{Z}$ which is homeomorphic to S^1. The function

$$\begin{aligned} f : \mathbb{R} &\to S^1 \\ t &\to (\sin(2\pi t), \cos(2\pi t)) \end{aligned}$$

so the universal property of the quotient gives a continuous bijection

$$\tilde{f} : \mathbb{R}/sim \to S^1.$$

Note any equivalence class of $\sim$ intersects $[0, 1]$ so using the continuous surjection

$$[0, 1] \xrightarrow{\text{inc}} \mathbb{R} \xrightarrow{\pi} \mathbb{R}/\sim .$$

This shows that $\mathbb{R}/\sim$ is the continuous image of a compact space hence it is comapct. But $S^1 \subseteq \mathbb{R}^2$ so it's Hausdorff hence

$$\tilde{f} : \mathbb{R}/\sim \to S^1$$

is a homeomorphism so $S^1 \cong [0, 1]/(0 \sim 1)$.

Example. Let $L = (\mathbb{R} \times \{0,1\})/((t,0) \sim (t,1))$ for all $t \neq 0$. We have seen this space is not Hausdorff so not a manifold. However it is locally Euclidean. Let π be the quotient map and $U = \pi((-1,1) \times \{1\})$. Note that $\pi^{-1}(U) = (-1,1) \times \{1\} \cup (-1,0) \times \{0\} \cap (0,1) \times \{0\} \subset \mathbb{R}$ is open so $U \subseteq L$ is open. Note that $\pi\mid_{(-1,1)\times\{1\}}: (-1,1) \times \{1\} \to U$ is a homeomorphism, so U is a Euclidean neighbourhood of the top origin.

Example. If M is a m-dimensional manifold and N is a n-dimensional manifold then $M \times N$ is a $(m+n)$-dimensional manifold.

Example. Consider the torus $T = S^1 \times S^1$ which is a 2-dimensional manifold. Using $S^1 = \mathbb{R}/\mathbb{Z}$ we have that $T = (\mathbb{R} \times \mathbb{R})/(\mathbb{Z} \times \mathbb{Z})$ i.e. the quotient of $\mathbb{R}^2$ by $x \sim y \iff x - y \in \mathbb{Z}^2$. Using $S^1 = [0,1]/(0 \sim 1)$ we get that $T = [0,1] \times [0,1] \approx$ where $(0,t) \approx (1,t)$ and $(s,0) \approx (s,1)$ for all $s,t \in [0,1]$. We can think of this geometrically as gluing two opposite edges of the unit square together to form a cylinder and then extending this cylinder and gluing end-to-end to form a doughnut shape which is what we usually understand as a torus as a subset of $\mathbb{R}^3$.

Example. Consider the equivalence relation $\sim$ on $[0,1] \times \mathbb{R}$ defined by $(0,t) \sim (1,-t)$. We can think of this as gluing on side "backwards" to the other side. This gives the set $M = [0,1] \times \mathbb{R}/\sim$. This is a 2-dimensional topological manifold which we will now prove.

Consider the subspaces $U_t = M \setminus L_t$ where $L_t = \pi(\{t\} \times \mathbb{R})$ where $0 < t < 1$. Note that $\pi^{-1}(L_t) = \{t\} \times \mathbb{R} \subset [0,1] \times \mathbb{R}$ hence L_t is closed since $\{t\} \times \mathbb{R}$ is closed. Note that

$$U_t = ([0,t) \times \mathbb{R} \ \sqcup \ (t,1] \times \mathbb{R})/\sim .$$

Doing the homeomorphism $(x,v) \longrightarrow (x,-v) \ \cdot \ (t,1) \times \mathbb{R} \xrightarrow{\cong} (t,1] \times \mathbb{R}$, defined as φ the relation $\sim$ translates to $\approx$ on $[0,t) \times \mathbb{R} \ \sqcup \ (t,1] \times \mathbb{R}$, namely

$$(0,s) \approx (1,s).$$

So $(\mathrm{id} \ \sqcup \ \varphi)$ gives a homeomorphism

$$([0,t) \times \mathbb{R} \ \sqcup \ (t,1] \times \mathbb{R})/\sim \xrightarrow{\cong} ([0,t) \times \mathbb{R} \ \sqcup \ (t,1] \times \mathbb{R})/\approx .$$

But this is homeomorphic to $(0,1) \times \mathbb{R}$ so $U_t \cong (0,1) \times \mathbb{R} \cong \mathbb{R}^2$.

As any $[x,y] \in M$ with $x \neq t$ for some $0 < t < 1$ we have that $[x,y] \in U_t$ a Euclidean neighbourhood. Any two points in M also lies in some U_t so Hausdorffness follows from Hausdorffness for Euclidean space.

Example. (Klein bottle) Combining the two examples, consider the relation $\approx$ on $[0,1]^2$ given by

$$(0,t) \approx (1,1-t) \quad \text{and} \quad (s,0) \approx (s,1).$$

The manifold $([0,1]^2)/\approx$ is the *Klein bottle*. The same consideration shows that this is a 2-dimensional manifold.

Example. (Connected-sum) If M, N are d-dimensional manifolds with a point $m \in M$ then we can find a Euclidean neighbourhood $m \in U$ with $\varphi : U \xrightarrow{\cong} \mathbb{R}^d$. If $n \in N$ then there is a Euclidean neighbourhood $n \in V$ with $\psi : V \xrightarrow{\cong} \mathbb{R}^d$. We then "glue" these two topological spaces together, with the *connected-sum* being the quotient space

$$M\#N = ((M \setminus \varphi^{-1}(B_1(0))) \ \sqcup (N\psi^{-1}(B_1(0))))/\sim .$$

Where we've defined $\sim$ as

$$\varphi^{-1}(v) \sim \psi^{-1}(v)$$

whenever $v \in \mathbb{R}^d$ has $|v| = 1$. This is covered by three opens sets,

$$\begin{aligned}
&M \setminus \varphi^{-1}\overline{B_1(0)} \\
&N \setminus \psi^{-1}\overline{B_1(0)} \\
&[(U \setminus \varphi^{-1}(B_1(0))) \ \sqcup \ (V \setminus \psi^{-1}B_1(0))]/\sim .
\end{aligned}$$

Using the induced homeomorphisms

$$\begin{aligned}
U \setminus \varphi^{-1}(B_1(0)) &\cong_\varphi \mathbb{R}^d \setminus B_1(0) \cong S^{d-1} \times (-\infty, 0] \\
V \setminus \psi^{-1}(B_1(0)) &\cong_\psi \mathbb{R}^d \setminus B_1(0) \cong S^{d-1} \times [0, \infty).
\end{aligned}$$

Hence the quotient is homeomorphic to $S^{d-1} \times \mathbb{R}$ which are both manifolds, so every point has a Euclidean neighbourhood. We can check by cases it's Hausdorff hence a topological manifold.

Example. Let $\mathbb{RP}^d = \{1\text{- dim subspaces of } \mathbb{R}^{d+1}\}$. Define a topology as a quotient of

$$\mathbb{R}^{d+1} \setminus \{0\} \text{ by } v \sim v' \iff \operatorname{span}(v) = \operatorname{span}(v')$$

which gives a topological space $\mathbb{RP}^d$.

Write $[x_0 : x_1 : \cdots : x_d]$ for the equivalence class of $(x_0, \ldots, x_{d+1}) \in \mathbb{R}^{d+1} \setminus \{0\}$ (so some $x_i \neq 0$). Thus the sets

$$A_i = \{[x_0 : \cdots : x_d] \mid x_i \neq 0\}$$

cover $\mathbb{RP}^d$. The preimage of A under the quotient map is

$$\{(x_0, \ldots, x_d) \in \mathbb{R}^{d+1} \setminus \{0\} \mid x_i \neq 0\}$$

which is an open subset, so A_i is open in $\mathbb{RP}^d$. The restriction of the quotient map $\pi \mid_{P_i}$ where $P_i = \{(x_0, \ldots, x_{i-1}, 1, x_{i+1}, \ldots, x_n) \in \mathbb{RP}^{d+1} \setminus \{0\}\}$. Then $\pi \mid_{P_i} : P_i \to A_i$ is a bijection. Also $P_i \cong \mathbb{R}^d$ by omitting the ith entry. We can see that $\pi \mid_{P_i}$ is easily checked to be a homeomorphism we can see that the A_i are Euclidean which cover the space.

To see that $\mathbb{RP}^d$ is Hausdorff let $p, q \in \mathbb{RP}^d$ with $p \neq q$. The group $GL_{d+1}(\mathbb{R})$ of linear isomorphism of $\mathbb{R}^{d+1}$ acts on $\mathbb{RP}^d$ continuously. We can find a matrix sending p and q to $\langle e_1 \rangle = \operatorname{span}(1, 0, 0 \ldots, 0)$ and $\langle e_2 \rangle = \operatorname{span}(1, 1, 0, \ldots, 0)$. It is now enough to show that we can separate $\langle e_1 \rangle$ and $\langle e_2 \rangle$. But these points are in $A_1 \cong \mathbb{R}^d$ so they can be separated by open sets. Hence $\mathbb{RP}^d$ is a d-dimensional manifold.

Any line in $\mathbb{R}^{d+1}$ is spanned by a unit vector, well-defined up to ± 1. We can also describe $\mathbb{RP}^d = S^d/(v \sim -v)$ which gives the same topology. This show that $\mathbb{RP}^d$ is the quotient of a compact space, hence it must be compact. Every equivalence class can be described by a point in the northern hemisphere $D^d_+ \subset S^d$. This is not unique, since if v is on the equator, then v is equivalent to another point in D^d_+. So $\mathbb{RP}^d = D^d/(v \sim -v)$ for $v \in \partial D^d$.

Theorem. A topological manifold is connected if and only if it is path connected.

Proof. We know that path connected impleis connected for all spaces already. Suppose M is a connected d-dimensional manifold. Then path component sof M partition into its disjoint

subspaces. It suffice to show that the path components are open as then they are compact closed. So this would disconnected the space if there was more than one of them.

Let $C \subseteq M$ be a path componenet and $c \in C$. Let $c \in U \cong \mathbb{R}^d$ be a Euclidean neighbourhood. Since $\mathbb{R}^d$ is path connected there is a path from any $c' \in U$ to c. Hence $U \subseteq C$. Thus C is open. $\square$

3.2 Homogeneity

We want to prove the following theorem.

Theorem. If X is a connected topological manifold and $p, q \in X$ then there is a homeomorphism $\varphi : X \to X$ such that $\varphi(p) = q$.

Remark. This is the same as saying the symmetry group of X acts transitively on X.

Example. Let $X = S^1 \sqcup \mathbb{R}$. Then there is no homeomorphism of X sending $p \in S^1$ to $q \in \mathbb{R}$. If there were, it would induce a homeomorphism $X \setminus \{p\} \xrightarrow{\cong} X \setminus \{q\}$. But $X \setminus \{p\}$ has two path-components and $X \setminus \{q\}$ has three path-components. So the conclusion of the theorem doesn't hold since we used a disconnected topological manifold.

Lemma. If $p, q \in B_1(0) \subset \mathbb{R}^d$ then there is a homeomorphism φ of $\mathbb{R}^d$ which sends p to q and is the identity outside $B_1(0)$.

Proof. Suppose *wlog* that $p = 0$. Consider first the homeomorphism $\rho : \mathbb{R}^d \to B_1(0)$ defined by

$$\phi(x) = \frac{x}{1 + |x|}$$

which has inverse

$$y \to \frac{y}{1 - |y|}.$$

There is a homeomorphism $x \to x + \rho^{-1}(q)$ of $\mathbb{R}^d$. When conjugated by ρ it gives a homeomorphism

$$\begin{aligned} \varphi'' : B_1(0) &\xrightarrow{\cong} B_1(0) \\ y &\longrightarrow \rho(\rho^{-1}(y) + \rho^{-1}(q) \end{aligned}$$

which sends $y = 0$ to q.

Define a function

$$\begin{aligned} \varphi' : \overline{B_1(0)} &\to \overline{B_1(0)} \\ y &\to \begin{cases} \varphi''(y) & |y| < 1 \\ y & |y| = 1 \end{cases}. \end{aligned}$$

We now aim to show that φ' is continuous. We will use the sequential definition of continuity. Let y_n be a sequence in $B_1(0)$ converging to $y_\infty \in \partial\overline{B_1(0)}$. So $|y_n| \to 1$. Hence $1 - |y_n| \to 0$ from

above. Now

$$\begin{aligned}\varphi'(y_n) &= \varphi''(y_n)\\ &= \rho(\rho^{-1}(y_n) + \rho^{-1}(q))\\ &= \frac{\frac{y_n}{1-|y_n|} + \frac{q}{1-|q|}}{1 + \left|\frac{y_n}{1-|y_n|} + \frac{q}{1-|q|}\right|} \to y_\infty = \varphi''(y_\infty).\end{aligned}$$

So φ' is continuous and a bijection since $\overline{B_1(0)}$ is compact and Hausdorff. So φ' is in fact a homeomorphism so defining

$$\begin{aligned}\varphi : \mathbb{R}^d &\to \mathbb{R}^d\\ y &\to \begin{cases}\varphi'(y) & |y| \leq 1\\ y & |y| \geq 1\end{cases}\end{aligned}$$

which is continuous on both closed sets, so continuous by the gluing lemma. The same applies to φ^{-1}. □

We're now ready to prove the theorem

Proof. Consider $x \sim y$ if and only if there exists a homeomorphism $\varphi : X \to X$ such that $\varphi(x) = y$. This is easily checked to be an equivalence relation, so partitions X into equivalence classes. As X is connected, it is enough to show that the equivalence classes are open, as having multiple equivalence classes would provide a disconnection for X. Let $x \in X$ and $x \in U$ a Euclidean neighbourhood, so there is some $\tau : U \xrightarrow{\cong} \mathbb{R}^d$. Let $C = \tau^{-1}\overline{B_1(0)}$, so $\mathring{C} = \tau^{-1}B_1(0)$. The lemma says that for each $q \in \mathring{C}$, there is a homeomorphism φ_q of $\mathbb{R}^d$ sending 0 to $\tau(q)$ and such that φ_q is the identity outside $\overline{B_1(0)}$.

Then $\tau^{-1} \circ \varphi_q \circ \tau$ is a homeomorphism of U sending p to q and is the identity outside of C. Define

$$\begin{aligned}\psi_q : X &\to X\\ x &\to \begin{cases}x & x \notin C\\ \tau^{-1}\varphi_q\tau(x) & x \in U\end{cases}\end{aligned}$$

which is well-defined since $\tau^{-1}\varphi_q\tau(x) = x$ in $U \setminus C$.

As X is Hausdorff, the compact subspace $C \subset U \subset X$ is closed, so $X \setminus C$ is open. So ψ_q is continuous by the open set version of the gluing lemma. Same applies for ψ_q^{-1}. Hence ψ_q is a homeomorphism of X sending $p \in X$ to $q \in \mathring{C}$. Hence the open neighbourhood $\mathring{C} \ni p$ lies in the equivalence class of p. So the equivalence class is open. □

3.3 Compact manifolds

Theorem. Let X be a compact d-dimensional manifold. Then there is a N such that $Y \subset \mathbb{R}^N$ with $X \cong Y$.

We'll look at a function before prove this theorem.

Consider

$$\begin{aligned}\sigma : \mathbb{R}^d &\to [0, \infty) \\ x &\to \begin{cases} 1 & |x| \leq 1 \\ 2 - |x| & 1 \leq |x| \leq 2 \\ 0 & |x| \geq 2 \end{cases}\end{aligned}$$

which is clearly continuous by the gluing lemma.

Proof. For $x \in X$ there is Euclidean neighbourhood $U_x \ni x$ and a homeomorphism $\tau_x : U_x \xrightarrow{\cong} \mathbb{R}^d$. Let $V_x = \tau_x^{-1} B_1(0) \subset U_x$, a smaller neighbourhood. Consider $\sigma_x = \sigma \circ \tau_x : U_x \to [0, \infty]$. Define

$$\begin{aligned}b_x : X &\to \mathbb{R}^d \\ y &\to \begin{cases} \sigma_x(y)\tau_x(y) & y \in U_x \\ 0 & y \notin U_x \end{cases}\end{aligned}$$

which is continuous again by the gluing lemma. Similarly we can consider $\sigma_x : X \to [0, \infty)$ by letting it be zero outside U_x which is also continuous by the gluing lemma. The open sets V_x cover X hence by compactness there is a finite subcover $V_{x_1}, \ldots, V_{x_n}$. Consider

$$\begin{aligned}e : X &\to (\mathbb{R}^d \times \mathbb{R})^n \\ y &\to (b_{x_1}(y), \sigma_{x_1}(y); \cdots ; b_{x_n}(y), \sigma_{x_n}(y))\end{aligned}$$

which is continuous by the definition of the product topology. Suppose $e(y) = e(y')$. Then $y \in V_{x_i}$ for some i, so $\sigma_{x_i}(y) = 1$. So $\sigma_{x_i}(y') = 1$ too so $y' \in V_{x_i}$. But then $b_{x_i}(y) = b_{x_i}(y')$, so $\tau_{x_i}(y) = \tau_{x_i}(y')$ hence $y = y'$ since τ_{x_i} is a homeomorphism so e is injective. So $e : X \to (X)$ is a continuous bijection from a compact space to a Hausdorff space, hence it is a homeomorphism. □

3.4 Surfaces

For this section we will just look at 2-dimensional manifolds which we call surfaces. We've already seen the torus, the Möbius band, the Klein bottle and $\mathbb{RP}^2$. We've constructed these by gluing sides of squares together and sometimes changing the direction of the gluing (in the Möbius band and Klein bottle). But there was nothing special about the square, more generally for a convex polygon $P \in \mathbb{R}^2$ having having $2n$ edges, each labelled by $a, b, c, \ldots$ such that every label that appears, appears exactly twice and each marked with an arrow.

Define an equivalence relation $\sim$ on P by identifying points on the two edges named α in the direction indicated. Formally, if the edges called α has vertices x, y with the arrow going $x \to y$ and the other α edge has vertices u, v with $u \to v$ then let

$$tx + (1-t)y \sim tu + (1-t)v$$

for all $t \in [0, 1]$ and all α.

Theorem. $P/\sim$ is a surface.

Proof. We'll just cover a sketch of the proof. First we'll show that $P/\sim$ is locally Euclidean. To show that $P/\sim$ is locally Euclidean for every equivalence class $[x]$ there are 3 cases.

(i) x is in the interior;

(ii) x is in the interior of an edge of P;

(iii) x is a vertex.

We know that (i) is obvious. For (iii) then suppose x is equivalent to other points $x', x'', \ldots$ for example. Then choose the radius ε open sector around each vertex $x', x'', \ldots$. Then in $P/\sim$, $[x]$ has a neighbourhood $(S \sqcup S' \sqcup S'' \sqcup \cdots)/\sim$. This is homeomorphic to the open ball cut up into sectors. This is homeomorphic to $\mathbb{R}^2$ by scaling angles on each sector. The proof of case (ii) follows similarly. This also provides a proof for Hausdorffness since we can pick a sector/disc around each point small enough. $\square$

The connected-sum of surface can be easily described through polygons in this way. We can consolidate edges together and reduce the number of vertices for shapes formed in this way.

Example. In a similar way we can obtain a 12-gon giving a connected-sum $\mathbb{RP}^2\#\mathbb{RP}^2\#\mathbb{RP}^2$ where all the edges are clockwise and read $xyxyststuvuv$. We can consolidate the edges to get the hexagon with clockwise edges reading $ppqqrr$. We can draw lines and cut and glue them to get the same image (TO BE COMPLETED NEXT LECTURE) as the connected-sum $T\#\mathbb{RP}^2$, hence we get that

$$\mathbb{RP}^2\#\mathbb{RP}^2\#\mathbb{RP}^2 \cong T\#\mathbb{RP}^2.$$

3.5 Triangulations

Every surface that we have discussed can be *triangulated* in the following sense.

Definition. (Triangle) A *triangle* in $\mathbb{R}^N$ with non-collinear points $pq, r \in \mathbb{R}^N$ is the space

$$\{t_0 : p + t_1 q + t_2 r \in \mathbb{R}^N : t_0 + t_1 + t_2 = 1, t_i \geq 0\}.$$

Remark. This formula gives the smallest convex set containing p, q, r by definition. The edges are the convex hulls of each pairwise choice of two vertices.

Definition. (Polyhedron) A collection $K = \{\Delta_1, \ldots, \Delta_n\}$ of triangles in $\mathbb{R}^N$ is called a *polyhedron* if $\Delta_i \cap \Delta_j$ is either empty or a common edge of both triangles. We set

$$|K| = \bigcup_{i=1}^{n} \Delta_i$$

as a subspace of $\mathbb{R}^N$.

Definition. (Triangulation) A *traingulation* of a topological space S is the data of a polyhedron K together with a homeomorphism

$$\phi : |K| \xrightarrow{\cong} S.$$